(a) Evidence-Preserving Annotation

Preserve Evidence

Problem x

Solution R

SaT spans → fixed steps

… … … … → s_1 s_2 s_3 …

Contiguous text spans — Numbered semantic steps

Preserve text · Fix boundaries

Attribute Errors via Consumed Content

Thin-film interference · real trace excerpt

Normal incidence: air / oil (n = 1.5, t = 500 nm) / water (n = 1.33)
Which wavelength is most strongly reflected? *Selected steps; text condensed*

s_2: Optical length = n × distance

Constructive interference: Δ = mλ — New error — omits reflection phase shift

s_3: $L = nt = 750\ nm$ — Correct

s_4: $2L = m\lambda$ — Propagated

s_6: $\lambda = 500\ nm \Rightarrow m = 3$ — Propagated

s_5: Smallest m ⇒ strongest reflection — New error (mixed) — *inherits wrong condition + adds invalid criterion*

Consumed content — Error-carrying dependency

An erroneous step can contain a valid output.

Construct Supervision

Structural consistency

Fully evaluable traces

Target for each step:

Correctness: correct / incorrect
Role: correct / new / propagated
Type: known new errors only

Recovery → correct role
Unknown type → type loss masked

(b) Hierarchical PRM Training

Problem + prefix + target step

Text-only student input

Shared Qwen3-4B backbone

h_i

Correctness → C / I
Error role → correct / new / propagated
Intrinsic type → 8 types

Targets + masks *(from a)*

Supervised targets

$$\mathscr{L} = \mathscr{L}_C + 0.5\mathscr{L}_R + 0.5\mathscr{L}_T$$

Type loss: known new errors only

(c) PRM Utilization

Process Diagnosis

s_1 s_2 s_3 …

PRM

Correctness
Origin or propagation
Intrinsic type

Weighted Answer Voting

s_1 … s_N

Shared candidate pool

Correctness scores → Group by answer + weighted vote → Selected answer